

RAG DATASET: COLLECTION & PREPARATION

Practical class notes for the Academic Question Answering System

Purpose: help you decide **what data to collect, where to get it, how to organize it, and how to prepare it** for a Retrieval-Augmented Generation (RAG) project.

This guide helps you answer	What you should produce
Which documents should I collect?	A source list and approved document folder
How should I organize them?	A consistent metadata sheet
How do I prepare documents for RAG?	Clean text, chunks, and source locations
How do I test the project?	A separate evaluation question set

Recommended beginner scope: Start with one subject or a small set of subjects, use trusted course materials, and expand only after the first end-to-end tests work.

1. What Does “Dataset” Mean in a RAG Project?

In machine learning, “dataset” often means a collection of examples used to train or evaluate a model. In a RAG system, the main dataset is usually a **knowledge collection**: documents the system searches at question time. RAG normally retrieves these documents; it does not require training a new language model on them.

You should plan for three related data collections

Collection	Contains	Main use
A. Knowledge documents	Syllabus, lecture notes, textbooks or open resources, lab manuals, official FAQs.	Provides evidence for answers.
B. Source metadata	Title, subject, unit, page, version, origin, permissions.	Makes search filters and citations possible.
C. Evaluation dataset	Test questions, expected key points, expected sources, review scores.	Measures whether retrieval, answers, and citations work.

For your project, keep the knowledge documents and evaluation dataset separate. Do not use only a list of question-answer pairs as the knowledge base if the goal is to answer questions from full academic notes.

2. What Types of Documents Should You Collect?

Priority	Document type	Why it is useful	Example metadata
High	Official syllabus	Defines subjects, units, and expected topic coverage.	Course, semester, academic year, unit.
High	Lecture notes / faculty notes	Main source of course-specific definitions and explanations.	Subject, unit, lecture, page.
High	Your own permitted study notes	Can add concise explanations and worked examples.	Author/creator, subject, topic, revision date.
Medium	Lab manuals and assignments	Supports practical, experiment, and assignment questions.	Course, lab/assignment number, topic.
Medium	Previous-year questions (PYQs)	Useful for testing question styles and topic coverage.	Year, course, unit, marks, question number.
Medium	Open educational resources or permitted textbook sections	Can add detailed explanations and examples.	Title, author, edition, chapter, license.
Optional	Official course FAQs / announcements	Supports course-specific logistics when in scope.	Issuer, date, source URL, version.

Suggested collection for a first prototype

- One current syllabus for the selected subject.
- Five to fifteen useful lecture-note or study-note files, if available and permitted.
- Relevant lab/manual material only if your system will answer practical questions.
- A small collection of PYQs for evaluation or coverage analysis.
- A separate set of questions written by you and checked against the notes.

These quantities are a manageable starting point, not a universal requirement. Quality, readable text, and traceable sources matter more than collecting a huge number of files immediately.

3. Where Can You Get the Data?

- **Your course:** official syllabus, faculty-provided notes, lab manuals, and permitted assignments.
- **Your own material:** notes you wrote yourself, after checking accuracy.
- **Open educational resources:** resources whose license allows your intended use; record the license and attribution.
- **Official documentation:** suitable when the project covers tools or technologies and those sources are relevant.
- **Past exam papers:** only from authorized or publicly available sources, and only if institutional rules permit their use.

Avoid collecting random internet PDFs without checking their source, accuracy, version, and permissions. Do not include private student records, answer sheets with personal details, or restricted exam material without authorization.

4. Make a Document Inventory

Before processing files, create a spreadsheet called **document_inventory**. Give each file a stable ID and record where it came from. This inventory becomes the source of truth for your project.

Field	What to record	Example
document_id	Unique stable identifier.	DBMS_NOTE_001
title	Readable source title.	DBMS Unit 2 Lecture Notes
subject	Course or subject.	DBMS
unit	Unit/chapter if applicable.	Unit 2
document_type	Syllabus, notes, lab manual, PYQ, etc.	Lecture notes
source_origin	Where it came from.	Faculty-provided course portal
source_date/version	Publication, semester, or revision date.	Semester 5, revised Aug 2026
file_name	Original filename.	dbms_unit2.pdf
page_count	Number of pages if known.	28
text_quality	Searchable PDF, scanned, mixed, etc.	Searchable PDF
permission/license	Permission or license notes.	Course use permitted
status	Pending review, approved, processed, excluded.	Approved

Simple file organization

Keep a clean folder structure, for example: **dataset/original_documents/**, **dataset/processed_text/**, **dataset/metadata/**, and **dataset/evaluation/**. Keep originals unchanged so you can trace a processed passage back to its source.

5. Check and Clean Your Documents

- Confirm that each file opens and is readable.
- Check whether the PDF contains selectable text or only scanned page images.
- For scanned files, OCR may be needed; inspect the output manually.
- Preserve headings, page boundaries, tables, formulas, and numbered steps when they matter.
- Remove repeated headers/footers only if doing so will not remove meaningful information.
- Mark duplicate files and choose the intended version.
- Check that extraction has not changed symbols such as $\leq$, $\geq$, subscripts, equations, or code snippets.
- Keep a record of files excluded and the reason for excluding them.

Do not silently “correct” academic content during cleaning. If a source appears to contain an error, flag it for review or keep the source faithful and record the issue.

6. Chunk-Level Dataset and Metadata

After extracting text, divide each document into chunks—small passages that can be retrieved independently. A chunk should usually focus on one coherent idea while keeping enough context to understand it.

Chunk field	Purpose
chunk_id	Unique ID for this passage.
document_id	Connects the chunk to the original inventory record.
chunk_text	The actual text passage used for retrieval and context.
subject / unit / section	Supports filtering and understandable citations.
page_start / page_end	Lets users find the passage in the original file.

Chunk field	Purpose
chunk_index	Records the chunk's order within the document.
document_version	Helps prevent citations to outdated material.

How to decide chunk size

There is no single correct chunk size for every dataset. Start with headings and paragraphs as natural boundaries, keep related explanation together, and test whether your retrieval results contain enough context. Overlap neighboring chunks a little if splitting would otherwise break a definition from its explanation.

Important distinction

The original document is the source. Extracted text and chunks are processed representations of that source. Preserve the mapping from each chunk back to its document and page or section; otherwise, citations become difficult to trust.

7. Build a Separate Evaluation Dataset

Your evaluation dataset is a set of questions used to test the system. Each question should have a reference answer or key points and the source passages that support them.

Field	Example
question_id	Q_DBMS_001
question	What is normalization in DBMS?
question_type	Definition / explanation
expected_key_points	Purpose of normalization; reduction of redundancy; relevant caveat if covered.
expected_document_id	DBMS_NOTE_001
expected_page_or_section	Unit 2, Normalization section, page 14
difficulty	Easy / medium / hard
answerable_from_dataset	Yes
review_status	Checked against source

Include several question types

- Definition: "What is a primary key?"
- Explanation: "Explain normalization in simple terms."
- Comparison: "Differentiate a process and a thread."
- Multi-part: "List and explain the ACID properties."
- Exact lookup: "Which topics are in Unit 3?"
- Paraphrased: ask the same concept using different wording.
- Out-of-scope: ask about a topic that is not in the collection.
- Ambiguous or conflicting: check whether the system asks for clarification or identifies conflicting sources.

Keep some test questions aside until development is complete. If you repeatedly tune the system using every test question, the final evaluation may overstate performance.

8. Dataset Quality Checks

Quality check	Question to ask	What to do if it fails
Relevance	Does this document match the project's academic scope?	Exclude it or label its scope clearly.
Authority	Can you identify who produced it?	Find a more reliable source or flag uncertainty.
Currency	Is it the intended version for this course?	Record version and resolve duplicates.
Readability	Can text be extracted accurately?	Use OCR or another extraction method and review.
Coverage	Are key syllabus topics represented?	Collect missing permitted material.
Traceability	Can each chunk be mapped to a source location?	Fix metadata before indexing.
Permissions	Can the project use and show this material as intended?	Get permission or exclude it.
Duplicates	Are multiple copies of the same source indexed?	Deduplicate or identify the canonical version.

9. Sample Dataset Plan for a College Project

Example scope: a prototype for one subject, such as DBMS or Operating Systems. The exact subject should match the materials you can legitimately access.

Dataset item	Initial plan	Reason
Syllabus	1 current file	Defines scope and units.
Lecture/study notes	A small curated set	Main answer evidence.
Lab/manual files	Only if practical questions are in scope	Adds relevant procedural information.
PYQs	A selection from permitted papers	Helps create realistic evaluation questions.
Evaluation questions	A varied set written and source-checked by you	Measures retrieval and answer quality.
Out-of-scope questions	Several questions absent from the collection	Checks whether the system avoids guessing.

Do not claim a particular dataset size is sufficient without testing. Start small, check retrieval and citation quality, and add documents based on observed coverage gaps.

10. Ready-to-Use Dataset Checklist

- I have defined the subject(s) and the kinds of questions the system should answer.
- Every document has a stable ID and a recorded source.
- I have checked accuracy, version, readability, and permissions.
- Original files are preserved separately from processed text.
- Each chunk stores its document ID and page/section metadata.
- I have checked extracted text for broken formulas, tables, and headings.
- My evaluation questions have reference key points and source locations.
- Some test questions are kept separate from development examples.
- I have questions with missing evidence to test abstention.
- I can trace any displayed citation back to the original source.

11. What to Explain in Your Report

Describe the dataset sources and why they were selected, the subjects and topic coverage, the number and types of documents actually collected, the cleaning and extraction process, the chunking strategy, the metadata fields, and how the evaluation questions were prepared. Report the real counts after collection—do not invent numbers in advance.

Example methodology wording: "The knowledge collection was built from approved academic materials. Each document was assigned a unique identifier and metadata describing its subject, unit, version, and source. Text was extracted and reviewed,

then divided into passages while preserving source locations. A separate evaluation set was created with questions, expected key points, and supporting source references.”